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Circadian Clock-Driven Inter-Organ Crosstalk in Cardiovascular-Kidney-Metabolic (CKM) Syndrome: Molecular Mechanisms, Pathological Progression, and Chronotherapeutic Strategies

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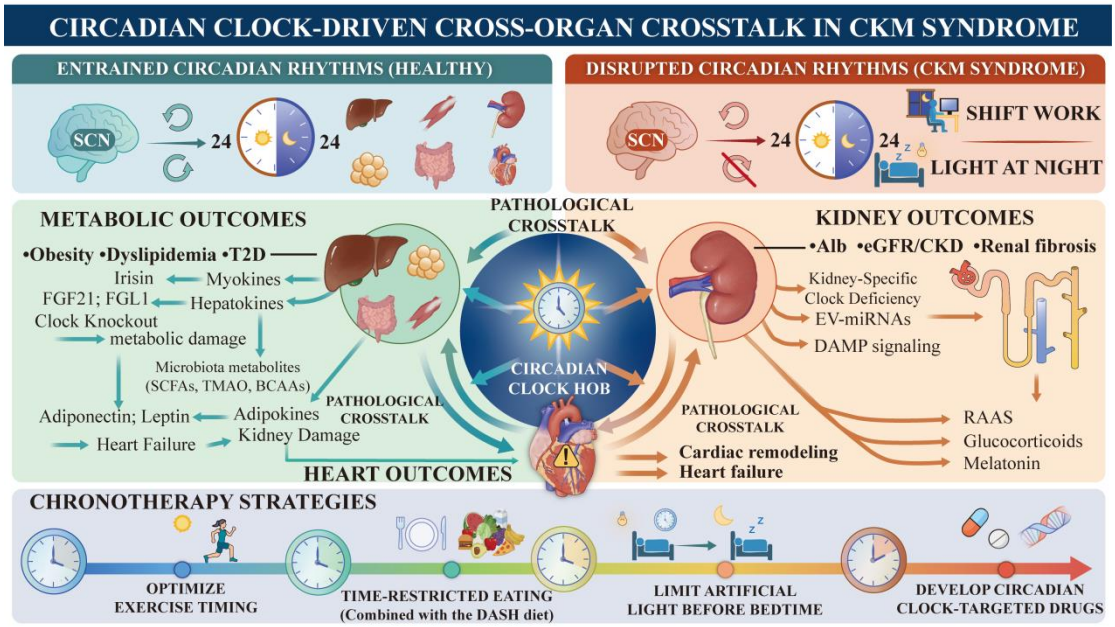
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Abstract Cardiovascular-kidney-metabolic (CKM) syndrome is not merely the simple superimposition of isolated target organ lesions, but rather a systemic disorder sustained by a complex network of inter-organ communication. The circadian clock functions as the central spatiotemporal regulator of physiological homeostasis, and the circadian disruption is increasingly recognized as an important contributor to CKM progression. In this review, we systematically elucidate the core mechanisms and biological functions of the circadian clock in mediating inter-organ crosstalk within CKM syndrome. We first delineate the peripheral clock architecture in the heart, kidneys, and crucial metabolic organs, followed by an in-depth analysis of the molecular mediators governing spatiotemporal communication among these organs. Building upon this framework, we further dissect how circadian disruption drives the pathological transition from an early metabolic-inflammatory network imbalance to coupled cardiorenal vascular injury. Finally, we comprehensively summarize multidimensional intervention strategies, encompassing chronopharmacological optimization, lifestyle interventions, and small-molecule modulators targeting the core clock machinery. Ultimately, this review aims to offer a novel chronobiological perspective for decoding inter-organ pathological crosstalk in CKM syndrome, thereby providing a robust scientific rationale for developing circadian-aligned, precision, and multitarget synergistic therapies.

Keywords Circadian clock; Cardiovascular-kidney-metabolic (CKM) syndrome; Inter-organ crosstalk; Chronomedicine

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Graphical Abstract



Abbreviations

AS: Atherosclerosis

BAT: Brown adipose tissue

BCAAs: Branched-chain amino acids

BMAL1: Brain and muscle aryl hydrocarbon receptor nuclear translocator-like protein 1

CKD: Chronic kidney disease

CKM: Cardiovascular-kidney-metabolic

CLOCK: Circadian locomotor output cycles kaput

CRY: Cryptochrome

CVD: Cardiovascular disease

eGFR: Estimated glomerular filtration rate

FGF15/19: Fibroblast growth factors 15 and 19

FGF21: Fibroblast growth factor 21

FGL1: Fibrinogen-like protein 1

GCs: Glucocorticoids

HF: Heart failure

MAFLD: Metabolic dysfunction-associated fatty liver disease

MI: Myocardial infarction

miRNAs: MicroRNAs

mtDNA: Mitochondrial DNA

PER: Period

PGC-1 α : Peroxisome proliferator-activated receptor gamma coactivator 1-alpha

PPAR α : Peroxisome proliferator-activated receptor alpha

REV-ERBα/β: Nuclear receptor subfamily 1 group D members 1 and 2

ROR: Retinoic acid receptor-related orphan receptor

ROS: Reactive oxygen species

S1P: Sphingosine-1-phosphate

SCFAs: Short-chain fatty acids

SCN: Suprachiasmatic nucleus

T2DM: Type 2 diabetes mellitus

TGF- β : Transforming growth factor beta

TMAO: Trimethylamine N-oxide

TRE: Time-restricted eating

TRF: Time-restricted feeding

TTFL: Transcription-translation feedback loop

WAT: White adipose tissue

3D : three-dimensional

Nomenclature note: Gene symbols are italicized and presented according to species-specific nomenclature, whereas protein symbols are presented in non-italic uppercase form. For example, *Bmal1, Clock, Fgf21, Per,* and *Cry* denote rodent genes, whereas BMAL1, CLOCK, FGF21, PER, and CRY denote the corresponding proteins or protein families.

1 Introduction

Cardiovascular-kidney-metabolic (CKM) syndrome has emerged as a major global public health challenge[1]. Rather than a mere physical comorbidity of the heart, kidneys, and metabolic organs, CKM syndrome is fundamentally a systemic disease orchestrated by a complex inter-organ communication network[2]. Traditional clinical research and intervention strategies have largely been confined to static pathological changes within isolated target organs. However, the pathological progression among CKM organs is not simply a spatial extension of lesions, but rather the consequence of multi-organ homeostatic decoupling across the temporal dimension. With the deepening understanding of inter-organ crosstalk, there is an urgent need to introduce a systems biology perspective to decipher the homeostatic imbalance within this intricate network.

The circadian clock provides a crucial spatiotemporal network perspective for understanding physiological homeostasis. The mammalian circadian clock system comprises a central clock and peripheral clocks distributed across organs such as the heart, kidneys, and skeletal muscle[3]. This system not only drives the cell-autonomous rhythms of the transcription-translation feedback loop (TTFL) at the cellular level, but also optimally integrates the functions of spatially separated organs within a 24-hour cycle through systemic coordination[4]. For terminological clarity, in this review, we use "circadian disruption" as an umbrella term for abnormalities in circadian phase, amplitude, period, rhythmicity, or temporal alignment. "Circadian misalignment" refers to an inappropriate temporal relationship between endogenous circadian timing and environmental or behavioral cycles, whereas "circadian desynchronization" is used more specifically to describe the loss of stable phase coordination among central, peripheral, or organ-specific clocks. When evidence concerns a particular tissue or cell type without demonstrating altered phase relationships between oscillators, we use more specific descriptions, such as tissue-specific clock dysfunction, loss of rhythmicity, or altered clock-gene expression. Circadian misalignment triggered by modern lifestyle alterations (e.g., shift work, nighttime exposure to artificial light) is widely recognized as a major risk factor driving cardiometabolic diseases[5]. Circadian disruption promotes systemic chronic low-grade inflammation[6]. It is also associated with cardiovascular and kidney events and all-cause mortality in patients with CKM syndrome[7]. At the molecular level, circadian disruption manifests as a rhythmic decoupling in the release and sensing of inter-organ communication mediators (such as neuroendocrine hormones, tissue-specific organokines, and extracellular vesicles)[8]. The desynchronization of peripheral clocks across various organs leads to a spatiotemporal mismatch in the inter-organ communication signals

received by target organs, thereby disrupting physiological homeostasis and accelerating CKM progression[9].

Our recent work examined circadian disruption in cardio-cerebral, cardio-hepatic, and cardio-renal comorbidities from a cardiovascular disease-centered perspective[10]. The present review addresses the broader CKM continuum by integrating tissue-intrinsic clocks across major CKM organs with four classes of inter-organ mediators and by linking their temporal dysregulation to the transition from metabolic–inflammatory instability to sustained cardiorenal and vascular injury. It further evaluates phenotype-guided chronopharmacological, lifestyle, and clock-targeted strategies for CKM management.

2 Circadian Clock Architecture in CKM Organs

In addition to central orchestration, the heart, kidneys, and key metabolic organs (liver, adipose tissue, intestines, and skeletal muscle) have evolved tissue-specific peripheral clocks (Figure 1). Elucidating the internal molecular timekeeping mechanisms within each organ is a critical starting point for precisely decoding inter-organ crosstalk in CKM syndrome.

2.1 The Cardiac Clock

The cardiac clock relies on the canonical TTFL. Circadian locomotor output cycles kaput (CLOCK) and brain and muscle aryl hydrocarbon receptor nuclear translocator-like protein 1 (BMAL1) form heterodimers that bind to E-box elements and drive the transcription of period (*Per)* and cryptochrome(*Cry)* genes [11]. PER and CRY proteins then translocate to the nucleus and repress CLOCK:BMAL1 activity[12]. Concurrently, CLOCK:BMAL1 mediates a secondary feedback loop by regulating Nuclear receptor subfamily 1 group D members 1 and 2 (*Rev-erbα/β)* to stabilize the rhythmic oscillation of *Bmal1* [13].

All intrinsic and resident cells in the heart possess molecular clocks, working synergistically to maintain cardiac homeostasis. The cardiomyocyte clock governs cardiac metabolism, contractility, and electrophysiology. In cardiomyocytes, CLOCK:BMAL1 directly regulates the transcription of *Dgat2* (lipid synthesis), *Pik3r1* (insulin signaling), *Usp2* (protein turnover), and *Nampt* (NAD+ synthesis)[14], thereby maintaining the oscillatory activity of mitochondrial respiration[15]. Furthermore, the cardiac clock rhythmically drives *Hcn4*[16] (pacemaker current), *Scn5a* (sodium current)[17], *Kcnh2* (potassium current)[18], and *Cacna1c* (L-type calcium channel)[19] to regulate electrophysiology. Additionally, the cardiomyocyte clock modulates sensitivity to hypertrophic stress; stress during the late active phase is highly susceptible

to inducing more severe myocardial remodeling (involving the PI3K/AKT/mTOR/S6 pathway)[20]. Cardiomyocyte-specific deletion of *Bmal1* leads to age-dependent cardiomyopathy and even heart failure (HF)[21, 22].

Endothelial cells are among the most abundant cell types in the heart. Endothelial BMAL1 and CLOCK intervene in angiogenesis[23] and atherosclerosis (AS)[24] by regulating the cell cycle network (CCNA1/CDK1) and autophagy, respectively. The circadian rhythm of cardiac fibroblasts is co-regulated by microenvironmental mechanical cues[25] and REV-ERBα/β, the latter of which represses transforming growth factor beta (TGF-β)-driven pathological transdifferentiation into cardiac myofibroblasts[26]. The circadian clock of cardiac immune cells influences immune-vascular interactions in AS and myocardial infarction (MI) by orchestrating chemokines and adhesion molecules[27]. Donor hearts procured during the active phase (12 pm - 12 am) exhibit a markedly increased risk of post-transplant rejection and decreased survival rates due to heightened antigen presentation and lymphocyte activation[28]. Currently, research on the function of circadian clocks in non-cardiomyocyte cardiac cells remains relatively limited, and the majority of underlying mechanisms await further elucidation.

2.2 The Renal Clock

Rhythmic gene expression, regulatory targets, and physiological functions differ among renal cell types, including podocytes, tubular epithelial cells, and renal progenitor cells. Together, these cell-specific clocks form a multilayered renal timing network. In glomerular podocytes, CLOCK directly drives approximately 13.4% of genes (including the autophagy genes *Becn1* and *Atg12*) to exhibit a 24-hour rhythm. Podocyte-specific deletion of *Clock* induces autophagy defects, podocyte loss, and proteinuria[29]. Furthermore, glucocorticoids (GCs) can reset the podocyte clock and induce the rhythmic expression of filtration barrier genes (*Nphs1* and *Nphs2*), suggesting a potential chronobiological mechanism in the treatment of glomerular diseases[30]. The renal tubular clock is deeply involved in systemic energy metabolism. Tubule-specific deletion of *Bmal1* severely disrupts metabolic pathways such as NAD+ synthesis, fatty acid transport, the carnitine shuttle, and β-oxidation, resulting in mitochondrial dysfunction and an approximately 50% reduction in plasma carnitine levels[31]. Melatonin upregulates the long noncoding RNA nuclear paraspeckle assembly transcript 1 through BMAL1, thereby promoting renal tubular epithelial-cell proliferation and attenuating tubular injury in experimental membranous nephropathy[32]. Core rhythmic genes in renal progenitor cells are deeply implicated in renal injury repair. Following proximal tubule injury, the expression of *per1a* and *nr1d1*

is strongly induced; both maintain the renal progenitor cell pool and promote tubular regeneration by repressing Lima1a (a negative regulator of epithelial-mesenchymal transition)[33]. Various pathological states and environmental factors can lead to cell-specific circadian disruption in the kidneys. Non-obese type 2 diabetes mellitus (T2DM) rats with nephropathy exhibit rhythmic abnormalities in *Per1* and multiple other genes in the renal cortex, accompanied by blood pressure circadian disruption[34]. Mice with lupus nephritis similarly display widespread renal circadian dysregulation, which is only partially restored upon disease remission[35]. Exposure to the environmental toxin aristolochic acid directly abolishes the PER2::LUC rhythm in renal tubular epithelial cells and exacerbates renal injury in the context of *Bmal1* deficiency[36]. Taken together, the internal molecular clock of the kidney is not a single homogeneous system, but rather a functional module composed of diverse cell types. Its cell-specific regulatory architecture provides a crucial foundation for understanding the mechanisms of nephropathy and developing chronotherapeutic strategies.

2.3 The Metabolic Clock

2.3.1 The Hepatic Clock

The hepatic circadian clock exhibits robust cell-autonomous properties; even when signal input from the central suprachiasmatic nucleus (SCN) is blocked, hepatocytes can still maintain robust, phase-coherent oscillations[37]. This system is intricately woven by three-dimensional (3D) chromatin conformation, protein homeostasis, and biomechanical sensing, constituting a highly complex, multi-layered network.

The 3D genome of the mouse liver undergoes dynamic remodeling, with the topologically associating domains of rhythmic genes exhibiting periodic compartmental switching. Compared to downstream networks, the chromatin network of core clock genes is significantly more dynamic, achieving precise spatiotemporal regulation of transcription[38]. The hepatic clock is additionally modulated by cellular mechanosensing. Hepatocyte volume and total macromolecule content exhibit diurnal oscillations of 30%–40%. The periodic variation in cell volume induces mechanical stress, which drives the rhythmic remodeling of the F-actin cytoskeleton. This, in turn, regulates the transcription of core rhythmic genes via the myocardin-related transcription factor/serum response factor pathway, achieving circadian synchronization at the biomechanical level[39].

Physiologically, the hepatic clock coordinates local glucose, lipid, and drug metabolism and inflammatory responses across the feeding–fasting cycle[40]. It also

regulates systemic energy homeostasis through hepatokine secretion[41]. For instance, hepatocyte-specific deletion of *Bmal1* not only leads to the collapse of the local metabolic network but also alters the circadian rhythm of secreted proteins (such as 14-3-3η, encoded by the *Ywhah* gene). This alteration subsequently remodels the expression pattern of *Cebpa* in white adipose tissue (WAT), ultimately affecting systemic metabolism[42]. Due to its status as a central metabolic hub, hepatic clock decompensation is intimately associated with metabolic diseases such as metabolic dysfunction-associated fatty liver disease (MAFLD), obesity, and T2DM[43].

2.3.2 The Adipose Tissue Clock

The TTFLs in WAT and brown adipose tissue (BAT) exhibit distinct tissue-specific characteristics. Transcriptional and metabolic oscillations in BAT are highly dependent on the NAMPT-NAD^+ pathway, whereas WAT exhibits a much weaker dependence on this axis[44]. Furthermore, peripheral clocks are highly susceptible to remodeling by nutritional stress. A high-fat diet can drive the formation of *de novo* chromatin loops at lipid-responsive enhancers via C/EBPβ. This process antagonizes REV-ERBα and potently activates the fibroblast growth factor 21 gene (*Fgf21*), constituting the epigenetic basis for metabolic reprogramming[45].

Physiologically, the adipose clock not only drives the diurnal alternation of lipid mobilization, oxidation (e.g., peaking in the early evening), and storage[46] but also profoundly cross-talks with the local immune microenvironment. The core clock gene *Bmal1* drives the rhythmic nocturnal secretion of IL-17 by adipose-resident immune cells (type 2 innate lymphoid cells and γδ17 T cells). This signal regulates *de novo* lipogenesis via the IL-17RC receptor; its impairment triggers systemic metabolic and thermoregulatory imbalances[47, 48]. Additionally, specific rhythmic molecules serve as crucial hubs mediating endocrine disorders. In obese individuals, early growth response 3 (a negative regulator of adipogenesis), which is suppressed by cortisol, is significantly downregulated, serving as a critical node connecting central and peripheral metabolic dysfunction[49]. Moreover, *Bmal2* downregulation or polymorphisms are closely associated with obesity, T2DM, and adipose inflammation[50]. Notably, the intrinsic human chronotype and the timing of interventions are vital for adipose health. In individuals with an evening chronotype (night owls), *PER1* in visceral fat is abnormally elevated, accompanied by a massive accumulation of inflammatory factors such as IL-1β and monocyte chemoattractant protein-1[51]. Conversely, early time-restricted feeding (TRF) can reset peripheral metabolic rhythms, significantly ameliorating body weight and inflammatory status in obese mice under isocaloric conditions[52]. In summary, the adipose tissue clock integrates a complex network of epigenetics, post-

translational modifications, and immune-metabolic interactions. Targeting these rhythms (e.g., via TRF or NAD^+ modulation) provides a solid foundation for chronotherapeutics in metabolic diseases.

2.3.3 The Intestinal Clock

The intestinal clock is a quintessential food-sensing peripheral oscillator. TRF anchors intestinal epithelial rhythms and restores diurnal microbial functions, including bile salt hydrolase-mediated bile-acid deconjugation, thereby improving metabolic homeostasis[53]. Conversely, feeding during the habitual rest phase can shift digestive-system peripheral clocks without comparably resetting the SCN, thereby inducing central–peripheral circadian desynchronization in experimental models[54]. In maintaining intestinal mucosal homeostasis, BMAL1 converges with the Wnt, Notch, and Hippo pathways to orchestrate intestinal stem cell proliferation, and directly targets apoptotic genes (*Bax*, *Trp53*, *Bak1*) to drive rhythmic epithelial injury repair[55]. In the immune microenvironment, REV-ERBα inhibits *Rorγt* transcription and supports type 3 innate lymphoid cells tissue residency and IL-22 secretion. This pathway contributes to rhythmic protection against intestinal inflammation[56]. If the rhythm is dysregulated, it induces the abnormal differentiation of enteric glial cells via the NR1D1/NF-κB axis, consequently accelerating gastrointestinal dysmotility[57].

Notably, the intestinal clock also serves as a critical hub connecting the microecological network with host systemic rhythms. On the one hand, the gut clock drives microbiome oscillations. Epithelial BMAL1 targets the promoter of NADPH oxidase 1 and generates rhythmic reactive oxygen species (ROS) production, which shapes the abundance of taxa such as *Bacteroidetes*. *Bmal1* deletion disrupts this microbial rhythm[58]. On the other hand, the gut clock can regulate host rhythms in a bottom-up manner. Epithelial BMAL1 drives the expression rhythm of the amino acid transporter SLC6A19, temporally regulating glutamine absorption to remotely modulate the activity of hypothalamic glutamatergic neurons and the host's sleep-wake cycle[59]. Concurrently, the abundance of colonic microbial metabolites (e.g., valine) correlates with serum glucagon-like peptide-1 levels and the abundance of butyrate-producing bacteria[60]. Taken together, driven by multidimensional inputs from feeding signals, microbial metabolites, and gut-brain axis signaling, the intestinal clock establishes a highly synergistic network of dynamic defense and metabolic communication between the host and its microbiota.

2.3.4 The Skeletal Muscle Clock

The skeletal muscle clock is not merely a passive metabolic executor but rather a

central hub that integrates environmental cues, mechanical contraction, and endocrine communication. In terms of molecular architecture, the skeletal muscle clock similarly relies on the TTFL, within which *Bmal1* serves as an indispensable core driver. Muscle-specific deletion of *Bmal1* leads to complete circadian disruption, whereas the targeted restoration of *Bmal1* expression solely within skeletal muscle fibers improves systemic glucose tolerance and metabolic flexibility, while reducing systemic inflammation[61]. In addition to being regulated by light-dark and feeding-fasting cycles[62], exercise constitutes the most critical physiological and mechanical zeitgeber (time cue) for skeletal muscle. BMAL1 deficiency attenuates the exercise-induced transcriptional responses of *Nr4a3* and *Ppargc1a*[63]. Further studies indicate that time-of-day-specific resistance training induces distinctly different plasma metabolite profiles and exercise benefits via the BMAL1/PGC1α4-FNDC5/irisin axis[64].

Pathophysiologically, disruption of the muscle clock rhythm not only directly triggers impaired glucose tolerance and insulin resistance but also compromises muscle mass and contractile function[65]. In skeletal muscle-specific *Bmal1* knockout mouse models, animals prematurely exhibit features of sarcopenia and mitochondrial dysfunction. Exogenous chronobiological interventions (such as scheduled exercise or melatonin supplementation) can effectively alleviate these degenerative defects and remodel the muscle microenvironment[66]. Additionally, the muscle clock prevents premature muscle loss by regulating the ROR/REV-ERB balance to control nocturnal protein degradation[67]. In summary, the decompensation of the skeletal muscle clock is intimately associated with sarcopenia, systemic metabolic syndrome, and accelerated aging. Chronotargeted interventions directed at skeletal muscle (such as chrono-exercise prescriptions) not only help maintain local muscle health but also provide a highly promising chronotherapeutic strategy for inter-organ metabolic diseases.

2.4 Mechanisms and Significance of Inter-Organ Circadian Crosstalk in CKM Syndrome

The central circadian clock orchestrates the phase synergy of peripheral clocks in CKM organs via neural and humoral mechanisms[68]. Building upon this, various organs (the heart, kidneys, liver, adipose tissue, skeletal muscle, and intestines) construct an intricate secondary communication network relying on endocrine factors, metabolites, and neurotransmitters.

Regarding the heart-kidney rhythmic axis, the renal clock gene *Per1* drives sodium transport networks such as the epithelial sodium channel. Its rhythmic disruption leads to an abnormal sodium excretion cycle, triggering nocturnal hypertension or a non-

dipping blood pressure profile—independent risk factors for inducing HF[69]. Furthermore, in the state of chronic kidney disease (CKD), elevated serum retinol and its binding protein can aberrantly activate the CLOCK pathway in monocytes. This drives the robust secretion of inflammatory factors and their infiltration into myocardial tissue, revealing the role of the kidney-immune-heart axis in the clock-mediated amplification effect of target organ damage[70]. Within the heart-metabolic rhythmic axis, adipose tissue acts as a core endocrine organ that not only releases circulating adipokines but also directly intervenes in the cardiovascular system via local paracrine signaling from epicardial/perivascular ectopic fat. Adipose tissue can release exosomal *miR-276a/b* to targetedly regulate the cardiac *Timeless* gene, directly impacting heart rate and the temporal characteristics of systole/diastole[71]. The collapse of the adipose clock is directly associated with systemic insulin resistance and an augmented risk of cardiovascular-metabolic events[72].

Synergy among metabolic organs constitutes the cornerstone of energy allocation. The liver regulates lipid metabolism and the *Cebpa* rhythm in WAT by secreting 14-3-3η[42]. Concurrently, systemic glucose homeostasis relies on the cross-tissue coupling of the liver-muscle clocks. Restoring an isolated local clock is ineffective; the rhythm of the liver-muscle metabolic cycle (e.g., the glucose-alanine cycle) must be reconstructed and supplemented with feeding cues to rescue systemic metabolic dysfunction[73]. Additionally, the microbiome-gut-liver axis provides another crucial dimension for rhythmic dialogue. The intestinal clock influences the polyunsaturated fatty acid pool by regulating *Fads1/2*, thereby distally maintaining the transcriptional rhythm of hepatic *Srebf1*. Deletion of epithelial *Bmal1* attenuates the synchronization capacity of the hepatic clock to an adverse diet, triggering insulin resistance[74]. Furthermore, commensal bacteria-derived microbial extracellular vesicles, acting as novel peripheral zeitgebers, increase during the nocturnal fasting period and directly activate hepatic gluconeogenesis[75]. Collectively, these organ axes illustrate that physiological CKM homeostasis depends not only on intact local clocks but also on temporal alignment among organs.

3 Molecular Mediators of Clock-Driven Inter-Organ Crosstalk in CKM Syndrome

To improve clarity and clinical relevance, we classify the mediators of circadian inter-organ crosstalk into four groups. The first three comprise neuroendocrine and autonomic signals, tissue-derived organokines, and circulating or microbiota-derived metabolites. The fourth includes extracellular vesicles and mitochondrial damage-associated molecular patterns that transmit epigenetic, inflammatory, and mitochondrial stress across organs. This framework emphasizes the major modes of inter-organ

communication and their links to CKM phenotypes rather than cataloguing individual signaling pathways.

3.1 Centrally Driven Neuroendocrine Rhythms

3.1.1 Canonical Central Rhythmic Hormones: Glucocorticoids and Melatonin

The stress-adaptive orchestration by GCs and the homeostatic restorative effects of melatonin complement each other to jointly coordinate inter-organ crosstalk within the CKM network. Under physiological conditions, pulsatile GC signaling activates adipose tissue to upregulate adiponectin, which subsequently triggers the CAMKK2-AMPK pathway in skeletal muscle, thereby achieving adipose-muscle metabolic synergy[76]. Under stress conditions, GC receptors and their regulatory factors exhibit temporal and tissue-specific expression remodeling across the heart, kidneys, adipose tissue, and muscle[77]. However, this communication mechanism is frequently disrupted during the progression of CKM syndrome. Patients with CKD universally exhibit rhythmic dysfunction of the hypothalamic-pituitary-adrenal axis[78]. Crucially, the decline in estimated glomerular filtration rate (eGFR) is accompanied by reduced local 11β-HSD2 enzyme activity, resulting in the loss of GC inactivation protection. This leads to the aberrant activation of mineralocorticoid receptors, exacerbating cardiovascular damage and thus serving as a critical mediator of heart-kidney pathological crosstalk[79].

As a core nocturnal rhythmic signal, melatonin rhythmically regulates lipogenesis and thermogenesis[80]; the disruption of its rhythm induces adipose inflammation and systemic energy imbalance[81]. In the kidney, melatonin may support tubular repair and attenuate proteinuria and fibrosis partly through clock-dependent transcriptional regulation[32]. Melatonin may also attenuate renal injury and secondary hypertension by modulating gut microbiota-derived metabolites and nitric oxide signaling. These effects may contribute to gut–kidney–heart crosstalk[82].

3.1.2 Autonomic Nerves and Peripheral Effector Endocrine Axes

In addition to central rhythms, fluctuations in autonomic nerve tension synergize with local endocrine axes to jointly construct the peripheral effector network of CKM cross-organ crosstalk (Figure 2).

At the level of neural conduction, the biological clock maintains sympathetic/parasympathetic balance and target organ perfusion through the rhythm of catecholamine release[83]. Under pathological conditions, patients with HF and CKD often experience sympathetic overactivation and a loss of autonomic rhythms, which

exacerbates cardiac remodeling and renal interstitial fibrosis[84, 85]. In an HF model induced by aortic constriction, renal sympathetic denervation can significantly alter the renal pathological process induced by cardiac pressure overload, directly confirming the core conductive role of autonomic nerves in heart-kidney crosstalk[86].

At the peripheral endocrine level, the molecular clocks of core metabolic organs such as the liver can remotely remodel the functional rhythms of target organs by releasing systemic factors. Liver-derived signals may contribute to the temporal regulation of the intrarenal renin–angiotensin system. In two hepatocyte-specific clock-disrupted mouse models, loss of *Bmal1* or *Per1/Per2* altered the phase and expression of several intrarenal renin–angiotensin system components without abolishing most daily rhythms[87]. Moreover, the decompensated activation of the renin-angiotensin-aldosterone system rhythm is precisely the core mechanism driving myocardial and glomerular fibrosis and mediating secondary heart and kidney damage[88]. In summary, by synergistically phasing the aforementioned neuroendocrine signals and their receptor responses, the central and peripheral clock networks precisely drive substrate metabolism, signal communication, and system-level feedback loops among various organs within the CKM system. Clinically, neuroendocrine and autonomic rhythm disruption provides a direct bridge between circadian disruption and recognizable CKM phenotypes, including non-dipping hypertension, morning blood pressure surge, sympathetic overactivation, CKD progression, and adverse cardiovascular remodeling. These mechanisms also support the clinical value of ambulatory blood pressure monitoring, sleep–wake assessment, and individualized chronotherapeutic strategies in patients with CKM syndrome.

3.2 Tissue-Specific Organokine Networks

3.2.1 Hepatokines

The hepatic clock coordinates distant CKM organs through time-dependent hepatokine secretion. Circulating FGF21 responds to time-of-day-dependent metabolic stress, with cold exposure producing a greater evening increase[89]. Liver – adipose FGF21 signaling supports insulin sensitivity and lipid utilization[90], with peroxisome proliferator-activated receptor alpha (PPARα)-related pathways contributing to adipose metabolic remodeling[91]. However, in CKM pathology, abnormal liver-derived FGF21 can promote myocardial hypertrophy through endocrine – autocrine communication with cardiomyocytes[92]. Liver-derived fibrinogen-like protein 1 (FGL1) also mediates liver – kidney crosstalk by accumulating in the kidney and activating activin receptor-like kinase 5/TGF-β signaling, thereby exacerbating renal interstitial fibrosis[93]. Other

rhythm-regulated hepatokines, including FGF15/19[94] and 14-3-3η[42], further coordinate bile acid metabolism and liver-adipose transcriptional synchrony, but their clinical relevance in CKM syndrome remains to be further clarified.

3.2.2 Myokines

Skeletal muscle participates in CKM crosstalk through exercise- and clock-regulated myokines, particularly irisin. The BMAL1/PGC-1α4/irisin axis contributes to time-of-day-dependent exercise responses[64]. After myocardial infarction, reduced skeletal-muscle irisin signaling is accompanied by endoplasmic-reticulum stress and muscle atrophy, whereas exercise-associated restoration of irisin promotes autophagy and attenuates muscle remodeling[95]. Muscle contraction and energy status also modulate irisin secretion[96], and experimental and clinical studies associate irisin with improvements in hepatic metabolic stress[97, 98] and fibrosis[99]. Thus, irisin provides a plausible link between the muscle clock, exercise timing, and remote organ responses.

3.2.3 Adipokines

The adipose clock regulates endocrine communication with other CKM organs. Adipocyte AMPKα2 signaling contributes to time-dependent adipose – muscle communication[100], while circadian variation in adiponectin and leptin influences systemic metabolic homeostasis[72]. Adiponectin maintains skeletal muscle homeostasis through Peroxisome proliferator-activated receptor gamma coactivator 1-alpha and FoxO3-related pathways, whereas leptin dysregulation is associated with declining muscle function[101]. In end-stage renal disease, however, a higher leptin/adiponectin ratio has paradoxically been associated with lower mortality, probably reflecting the confounding effects of protein – energy wasting and inflammation[102]. Together, clock-regulated organokines link the liver, skeletal muscle, and adipose tissue to insulin resistance, renal fibrosis, myocardial remodeling, and muscle wasting (Figure 2). Because much of the causal evidence remains preclinical, their clinical interpretation nevertheless requires consideration of sampling time, adiposity, inflammation, nutritional status, and renal clearance.

3.3 Circulating Metabolites and Micro-ecological Small Molecules

3.3.1 Gut Microbiota-Derived Metabolites

Feeding rhythms shape microbial oscillations, while microbiota-derived metabolites can feed back on host peripheral clocks[103]. Accordingly, disruption of hepatic or colonic epithelial clocks alters microbial rhythmicity and metabolite production, with consequences for hepatic metabolism and intestinal barrier and

inflammatory homeostasis[104, 105].

Short-chain fatty acids (SCFAs) may modulate peripheral clock-gene expression through receptor-mediated and epigenetic mechanisms[106]. Human data associate circulating SCFAs with brown-adipose-tissue oxidative metabolism[107], while cell-based evidence suggests that they attenuate skeletal-muscle inflammation and improve insulin-stimulated glucose uptake[108]. SCFAs may also contribute to renal and blood-pressure homeostasis, although the evidence remains largely preclinical[109].

By contrast, trimethylamine N-oxide (TMAO) is a diet–microbiota–host co-metabolite influenced by dietary choline and fasting status[110]. In experimental gut–kidney models, elevated TMAO has been linked to impaired intestinal barrier integrity[111] and activation of inflammatory and profibrotic signaling[112]. Clinically, higher TMAO concentrations are associated with lower GFR and increased mortality in CKD, although these associations vary among populations[113]. SCFAs and TMAO therefore provide candidate links between feeding–circadian disruption and CKM phenotypes, but timed causal studies in humans remain limited.

3.3.2 Small-Molecule Lipids

Lipid-derived metabolites may convey information about the temporal distribution of metabolic fuels. Experimental chronodisruption abolishes hepatic metabolomic rhythms and increases acylcarnitine levels[114], whereas exercise remodels hepatic acylcarnitine handling and selected acylcarnitines can enhance hepatocyte mitochondrial respiration[115]. Acylcarnitines may therefore reflect or participate in time-dependent fuel redistribution, although their role in human CKM crosstalk remains uncertain.

Sphingolipid signaling is similarly context dependent. Skeletal-muscle ceramide remodeling contributes to lipid-induced insulin resistance[116]. In diabetic mice, sphingosine-1-phosphate（S1P） partially restores myocardial vascular homeostasis[117], whereas dysregulated S1P metabolism contributes to CKD-associated vascular calcification[118]. These findings connect sphingolipids with metabolic and vascular injury, but do not yet establish clock-dependent inter-organ signaling.

Bile acids more directly link behavioral timing with systemic metabolism. Circulating bile acids exhibit 24-hour rhythms in humans, and sleep deprivation disrupts their temporal coordination with the lipidome[119]. Elevated bile acids have also been associated with circadian sleep disturbance in chronic liver disease[120]. Bile acid receptors regulate metabolic and immune responses[121], while obesity disrupts the coordinated rhythms of bile-acid metabolism and insulin sensitivity[122]. Thus, the

microbiota–bile acid axis may connect circadian misalignment with metabolic dysfunction, although its clinical relevance in CKM syndrome requires confirmation.

3.3.3 Amino Acids and Their Derivatives

Branched-chain amino acids (BCAAs) metabolism links circadian timing with metabolic and renal phenotypes. Plasma BCAAs exhibit 24-hour rhythmicity and are altered in sleep disturbance, obesity, T2DM, and Cardiovascular disease (CVD)[123]. In experimental diabetic kidney disease, disrupted BCAA homeostasis promotes macrophage-associated inflammation and fibrosis, whereas enhanced BCAA catabolism attenuates renal injury[124]. These findings identify BCAA metabolism as a potential therapeutic target at the T2DM–CKD interface, although time-of-day-specific clinical evidence remains limited.

Alanine and glutamine also mediate gluconeogenic substrate exchange among skeletal muscle, the liver, kidneys, and intestine[125], and a muscle–adipose creatine axis may link exercise-responsive energy metabolism with thermogenesis[126, 127]. However, direct evidence that the latter pathways transmit circadian signals remains insufficient.

3.3.4 Energy and Nucleotide-Derived Signals

The circadian clock regulates ketogenesis through the PPARα network[128], whereas hyperlipidemia disrupts diurnal metabolic coordination across the gut–liver axis[129]. Serum uric acid may provide a clinically accessible marker: night-shift work is associated with a higher prevalence of abnormal uric acid levels[130], and febuxostat modifies circulating adipokines in overweight or obese adults with asymptomatic hyperuricemia[131]. Whether urate lowering improves cardiorenal outcomes or exhibits time-dependent efficacy remains unclear.

The NAMPT–NAD^+ axis may also integrate circadian and metabolic homeostasis. Obesity disrupts the circulating eNAMPT–hypothalamic NAD^+ rhythm, while renal tubular clock loss[31] and skeletal-muscle Bmal1 signaling[132] are linked to impaired NAD^+-dependent mitochondrial metabolism. These pathways are measurable candidate biomarkers and targets, but prospective timed-intervention studies are still needed.

3.4 Extracellular Vesicles and Their Genetic Information Transmission

3.4.1 Exosomal MicroRNAs and Epigenetic Communication

Extracellular vesicles can transmit regulatory information among CKM organs. Circadian disruption increases adipocyte-derived exosomal miR-22-3p, impairing skeletal-muscle insulin signaling; circulating miR-22-3p is also associated with insulin-

resistance-related measures in shift-working nurses[133]. Broader evidence implicates skeletal-muscle clocks and exosomal microRNAs (miRNAs) network[134] in systemic metabolic communication[135], although direct evidence of rhythmic exosomal release remains limited. Hypoxia-induced exosomal miR-34b-3p can also target clock-gene expression in vitro[136]. Thus, exosomal miRNAs are candidate biomarkers and therapeutic targets for circadian-related metabolic dysfunction, but their tissue origin, sampling-time dependence, assay standardization, and clinical utility in T2DM, CKD, and CVD require prospective validation.

3.4.2 Mitochondrial DNA

Circadian clocks and mitochondrial quality-control pathways are reciprocally linked, although direct evidence that mitochondrial DNA (mtDNA) acts as a time-encoded inter-organ messenger remains limited[137, 138]. Under pathological stress, liver injury-derived extracellular mtDNA can activate TLR9-dependent renal tubular inflammation, providing proof-of-concept for liver–kidney communication[139]. In uremic rats, parallel alterations in mtDNA content and mitochondrial quality-control proteins occur in the kidney and heart and are partly corrected by omega-3 supplementation[140]. These findings suggest that mtDNA and mitochondrial quality-control markers may have potential as biomarkers for cardiorenal injury. However, their tissue origin, circadian variation, disease specificity, and prognostic value in CKD, and CVD require prospective clinical validation.

4 Circadian Clock Disruption Drives the Progression of CKM Syndrome

4.1 Metabolic-Inflammatory Network Imbalance in the Risk Initiation Phase

In the risk-initiation phase, circadian disruption primarily manifests as metabolic instability, low-grade inflammation, and early vascular activation. The significantly elevated neutrophil-to-lymphocyte ratio and C-reactive protein-to-albumin ratio in long-term night shift workers support an association between circadian misalignment and systemic chronic low-grade inflammation[141]. Furthermore, these pro-inflammatory consequences may also extend to maternal–fetal programming: maternal circadian disruption during pregnancy reduces maternal docosahexaenoic acid levels, thereby impairing neonatal myeloid-derived suppressor-cell function and exacerbating inflammation through PPARγ-related mitochondrial metabolic reprogramming[142]. Notably, in overweight individuals with T2DM, leukocytes show discordance between the expression of core clock transcripts and proteins, accompanied by enhanced leukocyte–endothelial adhesion and a pro-inflammatory phenotype[143]. These findings provide a clinically relevant link between molecular clock dysfunction,

immune activation, and early vascular injury.

Recent device-based prospective studies further support the metabolic relevance of behavioral circadian disruption. In the UK Biobank, greater exposure to light at night was associated with a graded increase in incident T2DM, independent of major lifestyle and genetic risk factors[144]. Similarly, lower amplitude and greater fragmentation of objectively measured rest–activity rhythms predicted a higher risk of incident T2DM[145]. Together, these observations suggest that disrupted light exposure and rest–activity organization may identify metabolic risk that is not fully captured by conventional clinical variables.

Moreover, there is a bidirectional driving relationship between the circadian clock and insulin resistance. *Bmal1* downregulation triggers glucose and lipid metabolism disorders, whereas targeted *Bmal1* overexpression can significantly improve lipid metabolism and repair synaptic damage[146]. Mechanistically, circadian disruption impairs the temporal coordination of substrate handling and inflammatory signaling across adipose tissue, skeletal muscle, and the liver. One representative example is the aberrant release of adipose-derived exosomal miR-22-3p, which suppresses insulin signaling in skeletal muscle and promotes insulin resistance[133]. Rather than representing an isolated molecular pathway, this adipose–muscle signal illustrates how local clock dysfunction may propagate metabolic stress across organs. Persistent insulin resistance, dyslipidemia, adipose inflammation, and endothelial activation can subsequently reinforce one another, promoting the transition from early metabolic dysfunction to T2DM and vascular disease.

Circadian-targeted interventions demonstrate tremendous potential in reversing early metabolic abnormalities. Clinical evidence indicates that a 12-week, 10h time-restricted eating (TRE) regimen can significantly improve body weight, blood pressure, and glucose metabolism in patients with metabolic syndrome by re-establishing the feeding-fasting cycle[147]. More recently, a randomized trial demonstrated that adding a personalized 8–10h TRE intervention to standard care produced modest improvements in glycated hemoglobin, body weight, and abdominal adiposity over 3 months[148]. Combining anti-inflammatory approaches with circadian interventions such as TRF may help disrupt the cycle of circadian disturbance, metabolic dysfunction, and inflammation during early CKM syndrome.

4.2 Cardiorenal Vascular Damage Coupling in the Pathological Evolution Phase

With disease progression, these initially functional disturbances converge on sustained cardiorenal and vascular injury. Large-scale cohorts have confirmed that

circadian syndrome—collectively characterized by metabolic disturbances, short sleep duration, and depression—independently and significantly increases the risk of combined cardiorenal events[7], prevalent and incident CKD, and all-cause mortality[149]. Because this construct includes established metabolic risk factors, these findings should not be interpreted as direct evidence that molecular clock disruption independently causes CKD. Nevertheless, prospective rest–activity data provide complementary evidence: among older adults, lower interdaily stability was associated with an approximately 25% higher 5-year risk of incident CKD after adjustment for obesity, hypertension, diabetes, sleep quality, and other clinical factors[150].

Under physiological conditions, the eGFR exhibits diurnal fluctuations, peaking at the onset of the active phase. This healthy rhythm is entirely lost during the progression of CKD, indicating that circadian clock failure synchronizes with renal function deterioration[151]. Furthermore, a reduced relative amplitude and elevated intradaily variability of the rest-activity rhythm are independently associated with CKM progression and cardiovascular events. Incorporating the aforementioned circadian parameters into basic clinical models significantly strengthens the evaluation of their association with mortality outcomes[152]. Consistently, a device-based prospective study of more than 70,000 adults showed that highly irregular sleep was associated with an increased risk of major adverse cardiovascular events, and adequate sleep duration did not fully offset this excess risk[153]. These observations support sleep and rest–activity regularity as potential digital biomarkers of CKM risk, although they have not yet been validated as treatment targets.

Several interacting mechanisms may link renal chronodisruption to cardiovascular injury. The renal clock contributes to the temporal regulation of sodium excretion, glomerular filtration, the renin–angiotensin–aldosterone system, and blood pressure. Disruption of these rhythms favors nocturnal hypertension and a non-dipping blood-pressure pattern, which are associated with CKD progression, vascular injury, and HF[83]. In parallel, systemic inflammation, oxidative stress, endothelial dysfunction, and fibrosis may reinforce bidirectional communication among the kidneys, vasculature, and heart[154].

Experimental evidence further supports a potential causal contribution of environmental circadian disruption. In mice with CKD, repeated reversal of the light–dark cycle aggravated renal dysfunction, oxidative stress, ATP depletion, and tubulointerstitial fibrosis. Treatment with melatonin or β-nicotinamide mononucleotide attenuated several of these abnormalities and partially restored BMAL1-related metabolic homeostasis[155]. These findings provide mechanistic proof of concept, but the

efficacy and safety of such interventions in patients with CKD remain unestablished.

Collectively, circadian disruption may accelerate CKM progression by converting early metabolic and inflammatory dysregulation into sustained renal, vascular, and cardiac injury. The most immediate translational opportunities include identifying severe rhythm disruption through sleep assessment, 24h ambulatory blood-pressure monitoring, and wearable-derived rest–activity measures. Prospective intervention trials are nevertheless required to determine whether restoring circadian regularity can reduce hard renal and cardiovascular outcomes beyond conventional CKM risk-factor management.

5 Multidimensional Intervention Strategies for CKM Syndrome Targeting the Circadian Clock System

5.1 Toward Phenotype-Guided Medication Timing

CKM syndrome is particularly suited to chronopharmacological approaches because several of its key clinical phenotypes—including nocturnal hypertension, non-dipping blood pressure, dyslipidemia, and disrupted sleep–activity rhythms—exhibit marked temporal variation. Aligning drug exposure with periods of greatest pathophysiological vulnerability offers a potentially low-cost strategy to optimize established therapies without increasing medication burden. However, the clinical value of chronopharmacology lies not in prescribing a universal clock time, but in matching medication timing to drug properties, individual circadian phenotypes, and treatment goals.

Large cardiovascular outcome trials have clarified the limitations of a one-size-fits-all bedtime strategy. In the TIME trial, evening administration of usual antihypertensive medications did not reduce major cardiovascular outcomes compared with morning administration[156]. The subsequent BedMed randomized trial similarly found no difference in death or major cardiovascular events between bedtime and morning dosing, with comparable safety profiles[157]. These neutral average effects argue against routine bedtime dosing for all patients, rather than against chronopharmacology itself, and suggest that administration time should generally preserve adherence, tolerability, and patient preference.

A more personalized approach may nevertheless be clinically relevant in selected circadian phenotypes. Subgroup analysis of the TIME trial suggested that alignment between antihypertensive dosing time and individual chronotype (e.g., evening dosing for evening chronotypes) was associated with a lower risk of non-fatal MI[158]. Short-term trials also indicate that dosing time can influence specific blood-pressure

phenotypes. In the OMAN trial, bedtime administration of olmesartan/amlodipine produced a greater reduction in nocturnal systolic blood pressure and improved the nocturnal dipping pattern over 12 weeks without increasing nocturnal hypotension, although renal and cardiovascular events were not evaluated[159]. Bedtime administration of telmisartan can optimize nocturnal blood pressure control in patients with concomitant diabetes and hypertension, whereas morning dosing is more conducive to suppressing exercise-induced blood pressure elevation, suggesting that decisions must be tailored to clinical phenotypes[160]. Together with interindividual variation in circadian blood-pressure profiles and drug responses[161], these findings support the use of 24h ambulatory blood-pressure monitoring, nocturnal hypertension or non-dipping status, chronotype, adverse-effect susceptibility, and adherence to guide individualized dosing. They do not justify routine bedtime dosing for all patients with CKD, CVD, or T2DM.

Lipid-lowering therapy provides complementary evidence that chronopharmacological effects may be drug-specific. Because hepatic cholesterol synthesis increases during the night, evening administration of short-half-life statins can better align peak drug exposure with endogenous cholesterol production. A recent meta-analysis found greater reductions in total cholesterol, low-density lipoprotein cholesterol, and triglycerides with evening rather than morning administration of short-half-life statins, whereas the relevance of dosing time was less evident for long-half-life agents[162]. Nevertheless, current evidence is largely based on lipid endpoints, and whether statin timing modifies cardiovascular or kidney outcomes remains uncertain.

Overall, chronopharmacology should be regarded as a precision layer superimposed on guideline-directed CKM therapy rather than as a universally superior dosing schedule. Its major translational opportunity lies in moving from fixed morning-versus-bedtime comparisons toward phenotype-guided strategies informed by ambulatory blood pressure, chronotype, sleep–activity patterns, drug pharmacokinetics, and CKM stage. Future trials should preferentially enroll patients with nocturnal hypertension, non-dipping blood pressure, T2DM, CKD, or marked circadian irregularity and determine whether personalized medication timing improves eGFR decline, albuminuria, heart-failure hospitalization, and major cardiovascular events.

5.2 Circadian-Aligned Lifestyle and Environmental Interventions

The core of lifestyle and environmental interventions lies in utilizing exogenous zeitgebers to re-establish the circadian synchrony of light, sleep, feeding, and exercise, thereby delaying the progression of CKM syndrome (Figure 3).

Dietary Rhythms: Meta-analytic evidence indicates that TRE produces modest improvements in body weight, insulin concentrations, and lipid profiles[163]. Notably, early TRF (with the first meal before 10:30 AM) significantly outperforms late TRF in improving fasting blood glucose[164]. Furthermore, TRE can remodel the blood pressure circadian rhythm[165]; when synergized with a Mediterranean or Dietary Approaches to Stop Hypertension diet, its cardiovascular protective potential is further amplified[166]. In adults with metabolic syndrome receiving standard medical care, a randomized trial showed that a personalized 8–10h TRE regimen modestly improved glycated hemoglobin, body weight, and abdominal adiposity over 3 months[148].

Exercise Rhythms: The metabolic benefits of exercise are highly dependent on the physiological state of the organism. In healthy populations, morning exercise is more conducive to fat mobilization[167]. However, in patients with metabolic syndrome, the maximal fat oxidation rate during afternoon exercise (16:00–17:00) is significantly elevated by 20.6% compared to morning exercise[168]. A small crossover trial in patients with T2DM receiving metformin suggested that morning exercise combined with pre-breakfast metformin produced more favorable glycemic responses than other exercise–medication timing combinations[169]. This exercise timing paradox reveals the remodeling of metabolic rhythms by pathological states, suggesting that personalized chrono-exercise prescriptions must be tailored based on the patient's clinical phenotype. Additionally, optimizing the overall diurnal activity rhythm (e.g., reducing prolonged sedentary time in the afternoon) is a crucial component in mitigating CKM risks[170].

Light and Sleep Environment: Light is the most potent zeitgeber for the SCN. Sufficient daytime exposure to natural light can enhance the amplitude of peripheral clocks, significantly improving glucose homeostasis in T2DM patients[171]. Conversely, artificial light at night strongly suppresses melatonin secretion, disrupts the sleep-wake cycle, and induces metabolic disorders, posing a particular hazard to vulnerable populations with concomitant vascular pathologies[172]. Device-derived sleep regularity has also been associated with major adverse cardiovascular events, independently of average sleep duration[153]. Currently, phototherapy (e.g., specific wavelengths of green light-emitting diode light) has preliminarily demonstrated clinical translational potential in mitigating oxidative stress[173] and synchronizing central and peripheral clocks[174]. Cohort studies have confirmed that maintaining robust sleep-activity rhythms is the cornerstone for resisting extreme environmental perturbations, improving lipid profiles, and reversing adverse cardiorenal prognoses[152, 175].

However, most studies of TRE, exercise timing, and light-based interventions have been relatively short and have primarily evaluated surrogate metabolic or blood-

pressure outcomes rather than major cardiovascular or renal events. Their long-term effectiveness and feasibility may also vary according to chronotype, work schedule, CKM stage, frailty, and adherence, and therefore require validation in phenotype-stratified clinical trials.

5.3 Emerging Clock-Targeted Pharmacology: Translational Opportunities and Challenges

Direct pharmacological manipulation of the circadian clock represents a potentially high-leverage strategy for CKM syndrome because clock components coordinately regulate metabolic, inflammatory, mitochondrial, and fibrotic programs across multiple organs. Small-molecule modulators of REV-ERBs, RORs, and CRY proteins have improved metabolic and tissue-remodeling phenotypes in experimental models, demonstrating that the circadian system is pharmacologically tractable. However, most compounds remain preclinical, and their effects cannot always be attributed exclusively to the intended clock target. For example, SR9009 is widely used as a putative REV-ERB agonist, but REV-ERB-independent effects on cellular metabolism and proliferation complicate mechanistic interpretation[176]. Improved target selectivity, pharmacokinetics, and long-term safety are therefore required before clinical translation.

A major challenge—and an opportunity for precision therapy—is the tissue-specific action of core clock proteins. Taking the core transcription factor BMAL1 as an example: myocardial BMAL1 expression is downregulated in diabetic cardiomyopathy. Targeted *Bmal1* overexpression promotes its binding to the *Bcl2* promoter, which suppresses inositol 1,4,5-trisphosphate receptor activity to alleviate mitochondrial calcium overload, thereby reversing cardiac remodeling[177]. Conversely, high glucose induces the aberrant upregulation of BMAL1 in vascular smooth muscle. Specific *Bmal1* knockout or BMAL1 inhibition can block its non-canonical activation of the core osteogenic transcription factor runt-related transcription factor 2, thereby mitigating vascular calcification[178]. This target paradox suggests that future circadian modulators urgently require high organ specificity (e.g., myocardium-specific agonism versus vascular-specific inhibition) to circumvent conflicting organ responses induced by systemic administration.

A nearer-term translational route may involve exploiting target-cell rhythmicity to optimize established therapies. GCs can reset the podocyte clock and induce rhythmic expression of genes involved in the glomerular filtration barrier[30], suggesting that treatment timing could potentially be aligned with renal cellular sensitivity. Whether

this approach improves proteinuria, eGFR decline, or treatment toxicity remains unknown. Overall, clock-targeted pharmacology is evolving toward precision strategies integrating circadian phase, tissue specificity, and drug-delivery technology. Future studies should determine whether selective clock modulation or circadian timing of established therapies provides incremental benefits beyond guideline-directed treatment for T2DM, CKD, and CVD.

6 Conclusion and Perspective

CKM syndrome is not simply an additive manifestation of injury in isolated target organs; rather, it reflects a systemic dysregulation of cross-organ communication in both space and time, driven by circadian disruption. This review systematically summarizes the mechanistic framework underlying this process, spanning from intracellular molecular oscillators to body-wide communication networks mediated by neuroendocrine hormones, organ-derived factors, microbial metabolites, and extracellular vesicles/mtDNA (Figure 2). Viewed through a chronobiological lens, the pathological coupling among CKM-related organs is revealed to be not only the spatial spread of inflammation, but also an inevitable consequence of temporal desynchronization of multi-organ homeostasis. In this context, strategies such as chronotherapeutic optimization, behavior-based interventions, and small-molecule modulators designed to restore core circadian rhythms hold considerable promise for the upstream prevention and treatment of CKM syndrome.

Despite increasing mechanistic evidence linking circadian dysregulation to inter-organ communication in CKM syndrome, several limitations constrain causal interpretation. Much of the current evidence is derived from nocturnal rodents, tissue-specific clock-knockout models, or acute environmental perturbations that do not fully reproduce the chronic, heterogeneous, and multimorbid nature of human CKM syndrome. Moreover, many studies examine a single signaling molecule or one directional organ axis in isolation. In the clinical disease milieu, however, the heart, kidneys, liver, adipose tissue, intestine, and skeletal muscle are simultaneously exposed to competing neural, endocrine, metabolic, inflammatory, and vesicular signals. It therefore remains difficult to distinguish causal mediators from compensatory responses or nonspecific markers of established tissue injury.

Important measurement uncertainties also apply to the mediators discussed in Section 3. Circulating organokines and metabolites reflect not only secretion but also tissue uptake, metabolism, and renal clearance, and their concentrations are influenced by sampling time, feeding status, adiposity, inflammation, medication use, and kidney

function. A single measurement therefore cannot reliably distinguish increased production from phase shifts, reduced rhythmic amplitude, or impaired clearance. For microbiota-derived metabolites, dietary exposure and interindividual microbial variability create additional sources of heterogeneity. For extracellular vesicles and cell-free mtDNA, tissue-of-origin attribution, isolation and normalization procedures, circadian variation, and causal specificity remain insufficiently standardized. These limitations currently restrict their use as clinical circadian biomarkers or therapeutic targets.

Barriers are also evident in the translation of circadian interventions. The neutral average results of large antihypertensive timing trials argue against a universal bedtime-dosing strategy and support phenotype-guided rather than clock-time-only prescriptions. Behavioral interventions such as TRE, scheduled exercise, and light modification are practical, but most studies have been short and have focused on surrogate outcomes. Their long-term adherence and safety remain uncertain, particularly in patients with advanced CKD, HF, frailty, or shift-work exposure. In addition, core clock components may exert tissue- and cell-specific, and occasionally opposing, effects, making systemic clock-targeted pharmacotherapy susceptible to off-target or paradoxical consequences.

To address these challenges, future chronomedicine strategies will need to evolve from generalized approaches toward high-resolution, multidimensional personalization. In clinical practice, flexible wearable devices may enable dynamic monitoring of individual circadian parameters, such as rest-activity rhythm profiles, which, when combined with continuous metabolic monitoring, could support the construction of personalized circadian maps. On this basis, clinicians may more precisely integrate chronopharmacology, TRE, and scheduled exercise interventions into a comprehensive management framework encompassing medication, diet, physical activity, and light exposure. In basic research, there is an urgent need to move beyond static, single-time-point observations toward dynamic tracking of circadian networks at high temporal resolution. Deep integration of multi-omics approaches, including single-cell RNA sequencing and spatial transcriptomics, with time-series analysis will be essential for mapping the circadian heterogeneity of CKM target organs, particularly resident and immune cell populations within the cardiorenal microenvironment. At the same time, multidimensional profiling of the gut microbiome should be further incorporated to clarify how rhythmic microbial metabolites reshape host epigenetic and transcriptional programs. Taken together, decoding circadian cross-organ communication will deepen our understanding of systemic CKM pathogenesis. It may also guide precision chronotherapy that aligns treatment with endogenous rhythms and interrupts multi-

organ injury. Such advances may ultimately open a new era of chronobiology-guided prevention and management of CKM syndrome.

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Contributions

Yue Liu and Yanfei Liu conceived this topic and revised the manuscript. Yajie Cai drafted the initial version of the manuscript, created the manuscript figures. Yeyan Qin, Mengmeng Zhu and Xi Cheng revised the manuscript. All authors have read and approved the final version of the article.

Figure preparation

The figures were manually created by the authors using BioRender.com. No generative AI was used. The relevant BioRender publication licenses were obtained for all figures.

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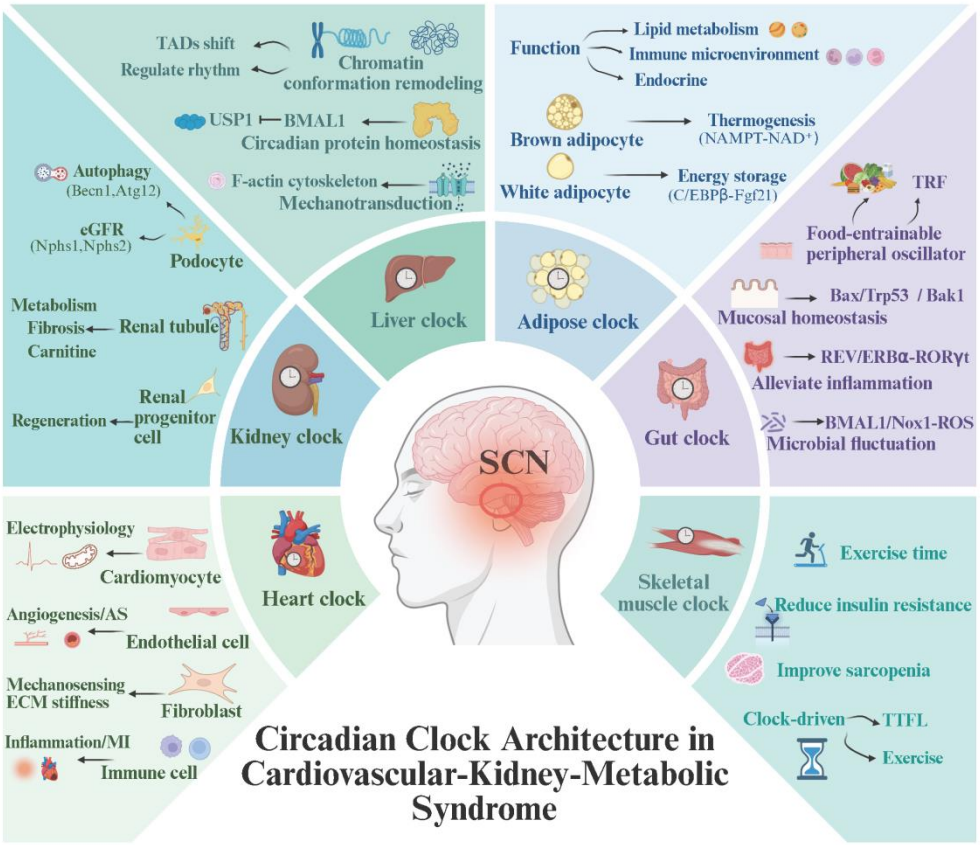


Figure 1. Circadian architecture of CKM syndrome. The circadian architecture of CKM syndrome is composed of the central clock, located in the suprachiasmatic nucleus, and peripheral clocks distributed across the heart, kidneys, and peripheral metabolic organs such as the liver, adipose tissue, gut, and skeletal muscle. In addition to being regulated by the central clock, each peripheral clock also possesses a robust autonomous oscillatory capacity. Abbreviations: AS, atherosclerosis; Atg12, autophagy-related 12; Bak1, B-cell lymphoma 2 antagonist/killer 1; Bax, B-cell lymphoma 2-associated X protein; Becn1, beclin 1; BMAL1, brain and muscle aryl hydrocarbon receptor nuclear translocator-like protein 1; C/EBPβ, CCAAT/enhancer-binding protein beta; ECM, extracellular matrix; eGFR, estimated glomerular filtration rate; F-actin, filamentous actin; Fgf21, fibroblast growth factor 21; MI, myocardial infarction; NAD^+, oxidized nicotinamide adenine dinucleotide; NAMPT, nicotinamide phosphoribosyltransferase; Nox1, nicotinamide adenine dinucleotide phosphate oxidase 1; Nphs1, nephrosis 1 (nephrin); Nphs2, nephrosis 2 (podocin); REV-ERBα, nuclear receptor subfamily 1 group D member 1; RORγt, retinoic acid receptor-related orphan receptor gamma t; ROS, reactive oxygen species; SCN, suprachiasmatic nucleus; TADs, topologically associating domains; TRF, time-restricted feeding; Trp53, transformation-related protein 53; TTFL, transcription–translation feedback loop; USP1, ubiquitin-specific peptidase 1.

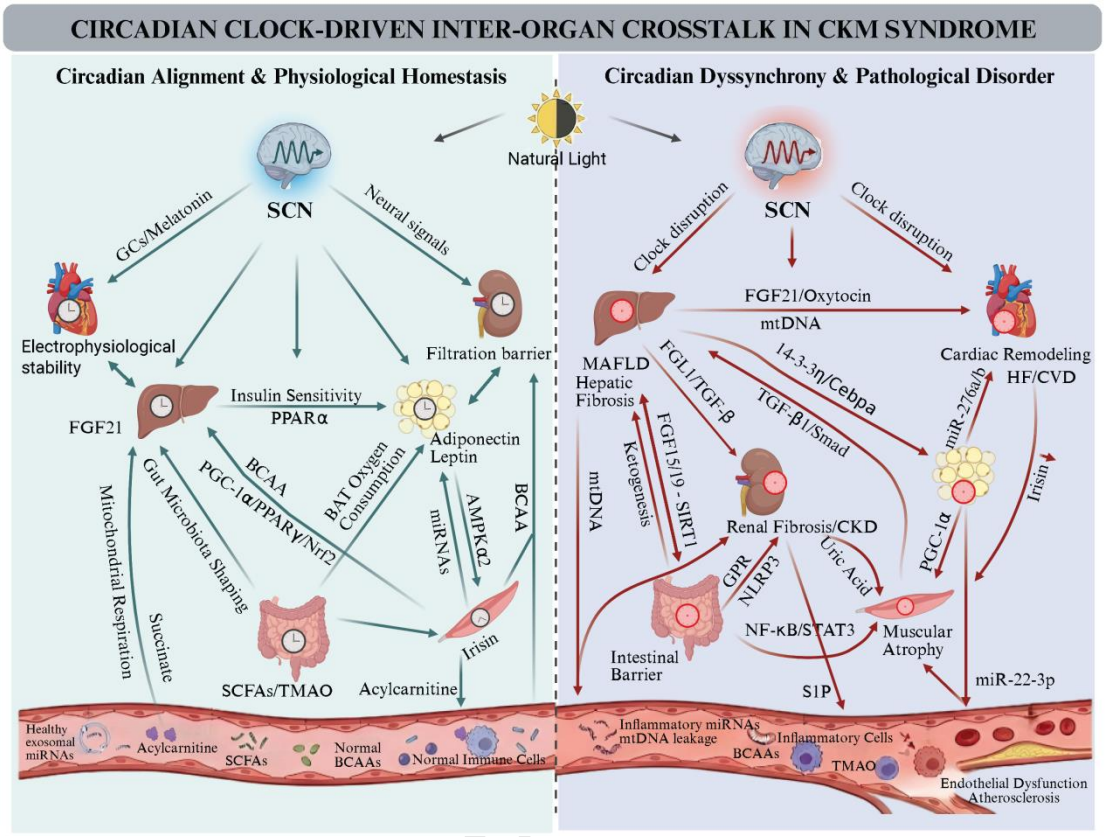


Figure 2. Mechanistic pathways of circadian clock-driven cross-organ crosstalk in CKM syndrome. The circadian system integrates photic signals via the central clock located in the suprachiasmatic nucleus to drive neuroendocrine rhythms and coordinate the phases of peripheral clocks. Concurrently, the circadian clock facilitates cross-organ crosstalk among CKM target organs (heart, kidneys, gut, liver, adipose tissue, and skeletal muscle) via neuroendocrine factors, organ-specific factors, metabolites, extracellular vesicles, and mtDNA to maintain systemic metabolic homeostasis. Abbreviations: AMPKα2, AMP-activated protein kinase alpha 2 catalytic subunit; BAT, brown adipose tissue; BCAAs, branched-chain amino acids; Cebpa, CCAAT/enhancer-binding protein alpha; CKD, chronic kidney disease; CKM, cardiovascular-kidney-metabolic; CVD, cardiovascular disease; FGF15/19, fibroblast growth factors 15 and 19; FGL1, fibrinogen-like protein 1; GCs, glucocorticoids; GPR, G protein-coupled receptor; HF, heart failure; MAFLD, metabolic dysfunction-associated fatty liver disease; miR-22-3p, microRNA-22-3p; miRNAs, microRNAs; mtDNA, mitochondrial DNA; NF-κB, nuclear factor kappa B; NLRP3, NLR family pyrin domain-containing 3; Nrf2, nuclear factor erythroid 2-related factor 2; PGC-1α, peroxisome proliferator-activated receptor gamma coactivator 1-alpha; PPARα, peroxisome proliferator-activated receptor alpha; S1P, sphingosine-1-phosphate; SCFAs, short-chain fatty acids;

SCN, suprachiasmatic nucleus; SIRT1, sirtuin 1; SMAD, SMAD family proteins; STAT3, signal transducer and activator of transcription 3; TGF-β, transforming growth factor beta; TMAO, trimethylamine N-oxide.

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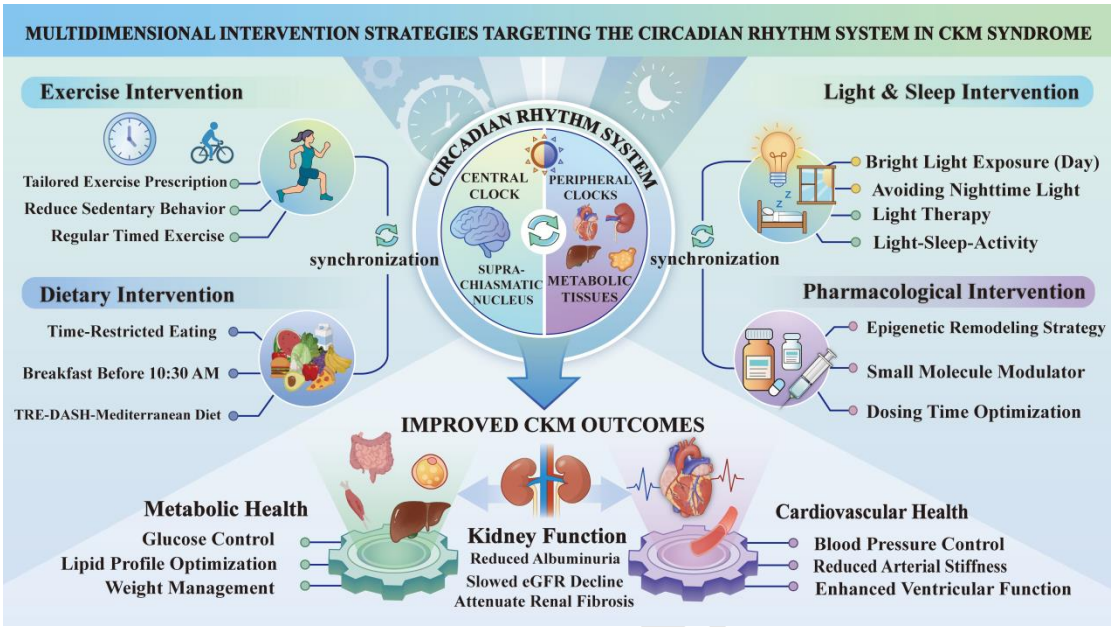


Figure 3. Multidimensional intervention strategies for CKM syndrome targeting the circadian clock system. Circadian-targeted interventions—encompassing the optimization of exercise timing, implementation of time-restricted feeding, reduction of artificial light exposure during nocturnal sleep, chronopharmacological optimization, and the development of small-molecule modulators targeting the circadian clock—hold significant promise for enhancing preventive and therapeutic efficacy. Abbreviations: AM, ante meridiem; CKM, cardiovascular-kidney-metabolic; DASH, Dietary Approaches to Stop Hypertension; TRE, time-restricted eating.

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Highlights

- Circadian clocks drive systemic inter-organ crosstalk in CKM.
- Specific organokines and metabolites mediate spatiotemporal CKM crosstalk.
- Exosomal miRNAs and mtDNA propagate pathological signals across CKM organs.
- Temporal decoupling shifts metabolic inflammation to cardiorenal injury.
- Chronotherapy provides precision multi-target CKM interventions.

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